AI Programming Foundations Project

Akhil Kumar

Machine Learning Researcher

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An Exploratory Data Analysis and Visualization of the AIRBNB housing data

This paper analyzes the housing patterns and trends for New York city. We use the 'New York City Airbnb Open Data | Airbnb listings and metrics in NYC, NY, USA (2019)' dataset obtained from Kaggle. The set of tasks performed are: load the data, analyze the columns and data properties, clean and transform it, perform visualizations and report the findings. The Jupyter Notebook for the project along with environment setup instructions is hosted on this Github link. [1]

Background

New York City is one of the most attractive tourist destinations, the Airbnb listings rent in the city is particularly complex and variable. Therefore, it is worth conducting research on the prediction of its rent and factors influencing it. Since 2008, guests and hosts have used Airbnb to expand on traveling possibilities and present a more unique, personalized way of experiencing the world. The dataset used describes the listing activity and metrics in NYC, NY for 2019. [3]

The core part of this research involves importing the data and making it ready for analysis and visualization. The aim is to spot patterns trends, clean up data and recommend best practices because it/they point out areas with high rates of properties and find housing patterns. The codebase can serve as a starting point for anyone wanting to train Machine Learning models and extract key insights from the real-world data.

Prerequisites

The dataset used in this research comes from the Inside Airbnb platform hosted on Kaggle. You can replicate and perform the analysis with a modern Mac/Windows device. It must have a working internet connection and Python3 pre-installed to set up the Jupyter Notebook.

The steps to install libraries and access the notebook are provided in the Github repository link. [1]

Analysis

The 2019 Airbnb (AB_NYC_2019) dataset seemed to be a very rich dataset with a wide range of columns that enabled us to conduct in-depth data exploration on each of the important columns that were provided. First we have identified hosts who make the most of the Airbnb platform and offer the most listings. Our top host for example has 327 listings.

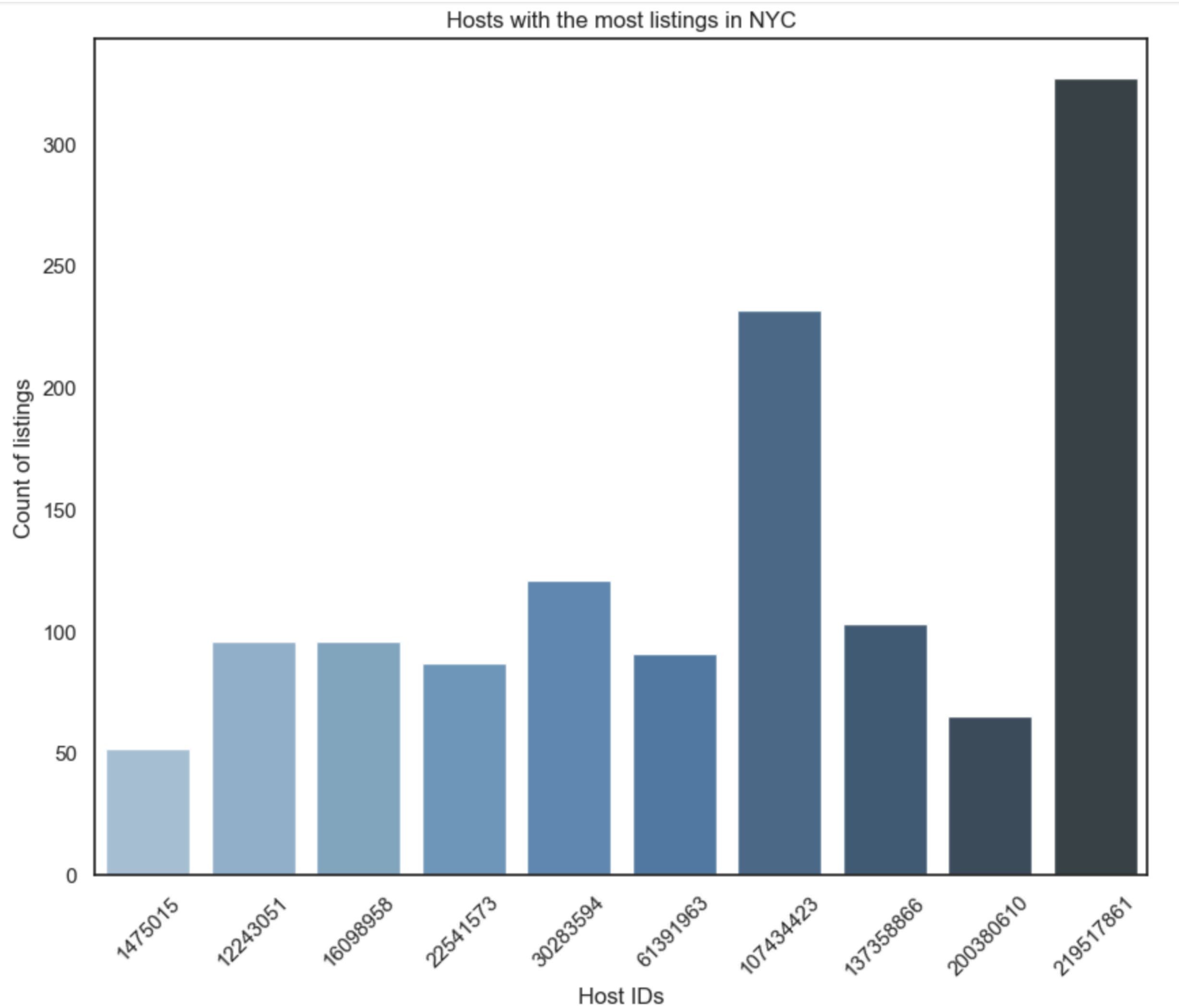

Figure 1: Top 10 hosts by number of listings

We then looked at listing densities by borough and neighborhood to determine which areas were more popular than others. Utilizing our latitude and longitude columns we then created a geographical heatmap that was color-coded according to the listing prices. In order to parse each title and examine current trends in listing naming as well as the number of most frequently used words by hosts we had to perform additional coding after returning to the first column containing name strings.

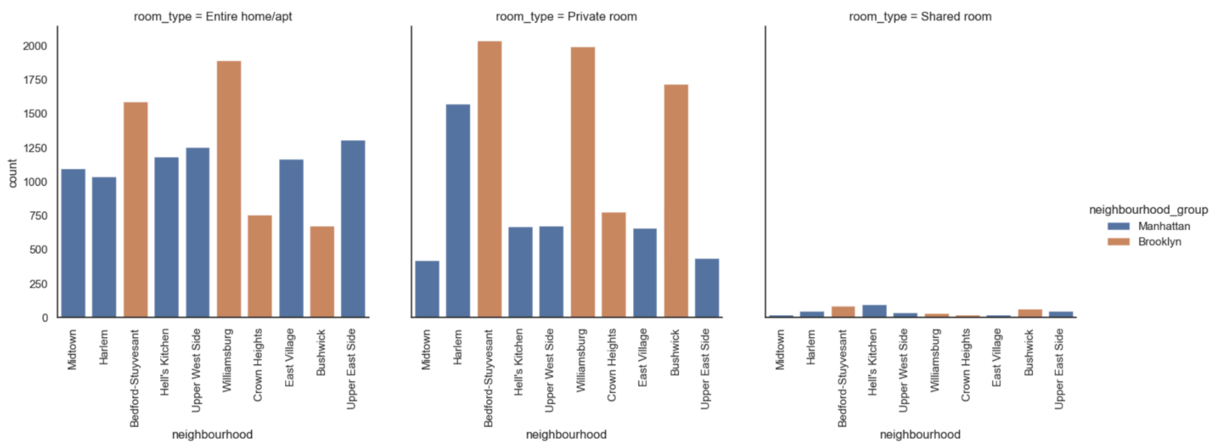

Figure 2: Counts for various home types and neighbourhoods

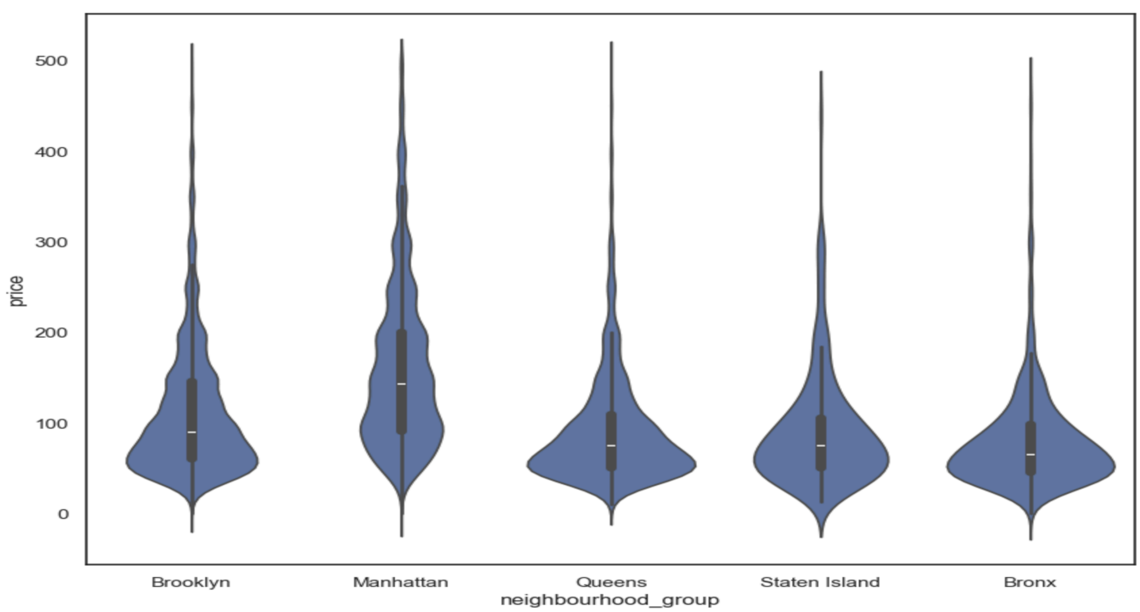

Figure 3: Density and distribution of prices for each neighborhood group

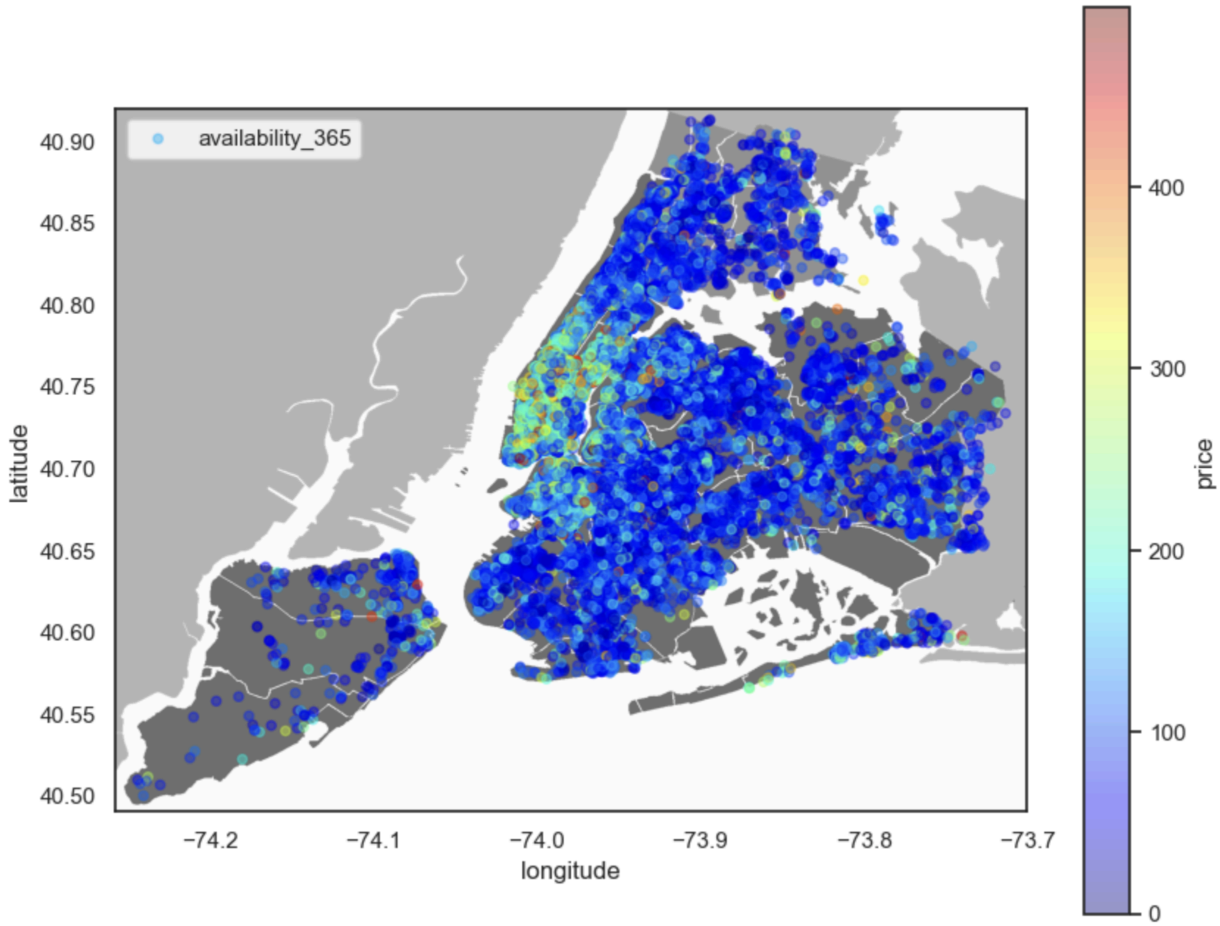

Figure 4: Visual representation of the property listings on the New York city map

Overall we were able to explain every step of the process and find a good number of intriguing relationships between features. For better business decisions platform control marketing campaigns, the introduction of new features and much more the Airbnb Data/Machine Learning team heavily imitates this data analytics at a higher level.

Data Cleaning, Bias and Outliers

While we try to be as objective as possible in picking the right columns to work on in the dataset, and what to remove, bias can still be present and the results or insights derived may vary for every person.

Missing data can cause problems in the analysis, such as biased results, incorrect statistical inferences, and errors in machine learning models. By identifying missing data, we handled it by dropping the rows with missing values. Another option here can be by imputing the missing values

Outliers are data points that are significantly different from other data points in a dataset. They can be much larger or much smaller than the other data points in the dataset and can distort the overall analysis of the data. We see this in the 'Room Type vs Price vs Neighbourhood Group' visualization above, where certain extremely high 'Private Room' and 'Entire Home/Apt' price values are pushing the average cost for these categories up, hence increasing bias. For ML training models, these outliers should be removed from the final datasets.

In our Jupyter Notebook, we demonstrate how to identify null value columns and either a. Drop the Rows OR b. Commute Z Scores and remove certain values below a certain threshold. We can also impute these values using statistical methods like average, median, mode, etc.

Conclusion

The following are some of the questions we were able to answer by conducting some intriguing analysis of the New York City Airbnb data during the project.

- What is the most popular type of room on Airbnb in New York City?

- Why do different types of rooms cost different amounts?
- Which areas of New York are the priciest to stay in?

Major data manipulation steps including data exploration cleaning analysis and visualization were undertaken in order to achieve these objectives. The following are the conclusions.

- Typical room types include private rooms and entire apartments.
- Compared to private and shared rooms hotel rooms and entire apartments are typically more costly.
- The most costly areas Manhattan and Brooklyn are home to more than 80% of the rooms.
- Certainly you will likely have to pay more if you wish to stay near the city's main attractions

References

[1] Github Link

https://github.com/akhil9650/Udacity-Capstone-Masters-in-AI/tree/main/AI%20Programming%20Foundations%20Project

[2] New York City Airbnb Open Data | Airbnb listings and metrics in NYC, NY, USA (2019). https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data

[3] Predicting New York City Rent through Machine Learning —— Based on Airbnb Data: Youzhe Gong. https://dl.acm.org/doi/10.1145/3749566.3749594